

CRYPTOGRAPHY & NETWORK SECURITY

Course Code: 23CS402 • Semester: 4-1 • Department of Computer Science & Engineering

Course Scope & Objectives: Classical and modern cryptographic ciphers, number theory foundations, symmetric encryption, public-key cryptography, digital signatures, hash algorithms, and network defenses.

Unit 1: Security Concepts & Classical Ciphers

Key Syllabus Topics: Security Goals (CIA Triad), Threat Models, Active vs Passive Attacks, Substitution Ciphers (Caesar, Playfair, Hill Cipher, Vigenere), Transposition Ciphers, Modular Arithmetic, Euclidean Algorithm, Extended Euclidean Algorithm.

Theoretical Framework & Concepts:

Cryptography protects data Confidentiality, Integrity, and Authenticity. Modular arithmetic provides finite algebraic structures where multiplicative inverses exist iff $\gcd(a, m) = 1$.

Implementation / Key Formula Reference:

```
// Modular Multiplicative Inverse: (a * x) % m == 1
```

Frequently Asked University Exam Questions:

- **Q1: Differentiate between active and passive security attacks.**

Solution: A: Passive attacks (eavesdropping, traffic analysis) monitor data without alteration; Active attacks (masquerade, replay, message modification, DoS) alter system resources or operations.

Unit 2: Symmetric Encryption (DES & AES)

Key Syllabus Topics: Block vs Stream Ciphers, Feistel Cipher Structure, Data Encryption Standard (DES), Triple DES (3DES), Advanced Encryption Standard (AES) Mathematical Operations (SubBytes, ShiftRows, MixColumns, AddRoundKey), Modes of Operation (ECB, CBC, CFB, OFB, CTR).

Theoretical Framework & Concepts:

AES uses substitution-permutation networks over Galois Field $GF(2^8)$. Cipher Block Chaining (CBC) XORs each plaintext block with previous ciphertext block using an Initialization Vector (IV).

Implementation / Key Formula Reference:

```
// CBC Encryption: C_i = Encrypt_K(P_i XOR C_{i-1})
```

Frequently Asked University Exam Questions:

- **Q1: Explain the 4 transformations in an AES round.**

Solution: A: SubBytes (non-linear S-Box substitution), ShiftRows (cyclically shifting row bytes), MixColumns (matrix multiplication in $GF(2^8)$), AddRoundKey (bitwise XOR with round key).

Unit 3: Asymmetric Cryptography & Key Management

Key Syllabus Topics: Fermat's and Euler's Theorems, Euler's Totient Function, Diffie-Hellman Key Exchange (Man-in-the-Middle Attack), RSA Algorithm (Key Generation, Encryption, Decryption, Security Proof), Elliptic Curve Cryptography (ECC).

Theoretical Framework & Concepts:

Public-key systems rely on trapdoor one-way functions. RSA security relies on the hardness of factoring large composite semiprimes $n = p * q$.

Implementation / Key Formula Reference:

```
// RSA: Choose primes p, q -> n=p*q, phi=(p-1)*(q-1)
// e*d = 1 mod phi -> Cipher = (M^e) mod n, Plain = (C^d) mod n
```

Frequently Asked University Exam Questions:

• **Q1: Perform RSA encryption for $p=7$, $q=11$, $e=13$, Message $M=9$.**

Solution: A: $n=77$, $\phi=60$. Since $13*37 = 481 = 1 \bmod 60$, $d=37$. Ciphertext $C = (9^{13}) \bmod 77 = 4$.

Unit 4: Cryptographic Hash Functions & Digital Signatures

Key Syllabus Topics: Cryptographic Hash Properties (Pre-image, Second Pre-image, Collision Resistance), Birthday Paradox, SHA-256 / SHA-512 Architecture, HMAC, Digital Signature Standard (DSS / DSA), PKI (X.509 Certificates, Certificate Authorities).

Theoretical Framework & Concepts:

Hash functions produce deterministic fixed-size digests. Digital signatures encrypt hash digests using the sender's private key, guaranteeing non-repudiation and integrity.

Implementation / Key Formula Reference:

```
// Digital Signature: Signature = Sign_PrivateKey(Hash(Message))
// Verification: Verify_PublicKey(Signature) == Hash(Message)
```

Frequently Asked University Exam Questions:

• **Q1: What are the security properties required for a cryptographic hash function?**

Solution: A: Pre-image resistance (one-way), Second pre-image resistance (weak collision), and Collision resistance (hard to find any pair $x \neq y$ with $H(x) == H(y)$).

Unit 5: Network & System Security Protocols

Key Syllabus Topics: IPsec Architecture (AH, ESP, Tunnel vs Transport Mode), SSL/TLS Protocol Stack & Handshake, PGP for Email Security, Firewalls (Packet Filtering, Stateful Inspection, Application Gateways), Intrusion Detection Systems (IDS), DDoS Mitigation.

Theoretical Framework & Concepts:

IPsec operates at Network Layer providing end-to-end IP packet encryption (ESP) and authentication (AH). Stateful firewalls track active TCP session states to block unsolicited packets.

Implementation / Key Formula Reference:

```
// IPsec Transport Mode: [IP Header | ESP Header | Encrypted Payload | ESP Trailer]
```

Frequently Asked University Exam Questions:

- **Q1: Differentiate between IPsec Transport Mode and Tunnel Mode.**

Solution: A: Transport mode encrypts only the IP payload (host-to-host); Tunnel mode encrypts the entire original IP packet and adds a new outer IP header (gateway-to-gateway VPN).